

D4 Interference

D4.1 Complete the question in the table:

Wavelength	Slit separation	Distance to screen /m	Fringe spacing /mm
500 nm	0.25 mm	1.50	(a)

D4.2 Complete the question in the table:

Wavelength	Slit separation	Order of interference n	Angle to 'straight through' direction
450 nm	0.050 mm	3	(a)

D4.3 White light shines normally onto a diffraction grating with 500 lines/mm. It then reaches a wall which is 1.50 m from the grating. The wall is perpendicular to the original beam of white light. A person who can see light with wavelengths between 380 nm and 750 nm measures the width of the first order spectrum on the wall.

- How wide is the first order spectrum?
- How wide is the gap between the first and second order spectra?

CHAPTER D. WAVES

Answers

$$\text{D4.1} \quad \frac{1.50 \times 5 \times 10^{-7}}{2.5 \times 10^{-4}} = 3.0 \text{ mm}$$

$$\text{D4.2} \quad \sin \theta = \frac{3 \times 4.5 \times 10^{-7}}{5.0 \times 10^{-5}} = 0.027 \quad \theta = 1.547^\circ$$

D4.3

$$\text{a) } \sin \theta_{380} = \frac{1 \times 3.8 \times 10^{-7}}{10^{-3}/500} = 0.190$$

$$\tan \theta_{380} = \frac{0.190}{\sqrt{1 - 0.190^2}} = 0.193525$$

$$\sin \theta_{750} = \frac{1 \times 3.8 \times 10^{-7}}{10^{-3}/500} = 0.375$$

$$\tan \theta_{750} = \frac{0.375}{\sqrt{1 - 0.375^2}} = 0.404520$$

$$\text{Width} = 1.50 \times (0.404520 - 0.193525) = 0.31649 \text{ m}$$

b) First order red:

$$\sin \theta_{750} = \frac{1 \times 3.8 \times 10^{-7}}{10^{-3}/500} = 0.375$$

$$\tan \theta_{750} = \frac{0.375}{\sqrt{1 - 0.375^2}} = 0.404520$$

Second order blue:

$$\sin \theta_{380} = \frac{2 \times 3.8 \times 10^{-7}}{10^{-3}/500} = 0.380$$

$$\tan \theta_{380} = \frac{0.380}{\sqrt{1 - 0.380^2}} = 0.410817$$

$$\text{Width} = 1.50 \times (0.410817 - 0.404520) = 9.4455 \text{ mm}$$